Nuclear physics practice

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1. Problem 1

We know that negative beta decay can occur if, for a nucleus with mass number A, the atomic number Z satisfies Z < Zmin(A), while electron capture can occur if Z > Zmin(A). Therefore, one would expect that no nucleus could undergo both negative beta decay and electron capture simultaneously. However, there are some isotopes, for example 40K that decays with a probability of 88 % by negative beta decay and with a probability of 12% by electron capture. Give an explanation to this phenomenon using the Weiszacker result for the isobaric energies (A=cst.). What could be the reason for the difference in the energies of the neighbouring isobars of 40K?

Solution:

If the isobaric masses lie on a single parabola, the nucleus cannot undergo both processes simultaneously. But for even mass numbers A, pairing effects split the isobars into two separate parabolas:

- one corresponding to even-even nuclei,
- one corresponding to odd-odd nuclei.

The pairing term in the Weizsäcker formula is

$$\delta(A,Z) = \begin{cases} +\delta_0 & \text{even-even nuclei,} \\ 0 & \text{odd } A, \\ -\delta_0 & \text{odd-odd nuclei.} \end{cases} \tag{1}$$

The isotope ^{40}K has

$$Z = 19, \qquad N = 21, \tag{2}$$

thus it is an odd-odd nucleus. Its neighbouring isobars are

$$^{40}\text{Ar}: \quad Z = 18,\, N = 22 \qquad \text{(even-even)}, \tag{3}$$

$$^{40}\text{Ca}: \quad Z = 20,\, N = 20 \qquad \text{(even-even)}. \tag{4}$$

Because even-even nuclei gain additional binding energy from pairing, both ^{40}Ar and ^{40}Ca are shifted downward in energy relative to the odd-odd nucleus ^{40}K.

As a consequence, the mass of ^{40}K is higher than both neighbouring isobars:

$$M(^{40}\text{K}) > M(^{40}\text{Ar}), \qquad M(^{40}\text{K}) > M(^{40}\text{Ca}). \tag{5}$$

Therefore both decay channels are energetically allowed:

$$^{40}\text{K} \to {}^{40}\text{Ca} + e^- + \bar{\nu}_e, \tag{6}$$

$$^{40}\text{K} + e^- \to {}^{40}\text{Ar} + \nu_e. \tag{7}$$

The reason for the different energies of the neighbouring isobars is thus the pairing term in the Weizsäcker formula. The even-even nuclei are more strongly bound, while the odd-odd nucleus ^{40}K is less strongly bound and therefore unstable against decay in both directions.

2. Problem 2

Consider the semi-empirical binding energy formula and discuss its terms. Now take an object that consists solely of neutrons.

a. Identify the remaining terms in the Weiszacker formula! Distinguish the attractive and repulsive parts.

In order for the object to remain bound, additional attractive terms must be included. One such possibility is when the number of neutrons (hence the mass of the object) is high enough for the gravitational force to dominate, in which case our nucleus resembles a neutron star!

b. Investigate the Weiszacker model for such a neutron star and utilize it to estimate the minimum mass number (A) and mass (M) for which a bound state can form. Consider homogeneous mass distribution and spherical symmetry to determine the gravitational potential energy term in the binding energy formula. (Use the fitting constants of the Weiszacker formula presented in the lecture.)

Solution:

The remaining terms in the binding energy formula:

$$E(A) = -e_V A + e_F A^{2/3} + e_a A - |E_g|. \tag{8}$$

Let us use the model in which our star is homogeneous and spherically symmetric. The magnitude of the gravitational force arising between two masses can be calculated from Newton's law:

$$F = G\frac{m_1 m_2}{r^2}, \tag{9}$$

where m_1 and m_2 are the masses of the bodies, r is the distance between them, and $G = 6.67{\cdot}10^{-11}\,\mathrm{Nm}^2/\mathrm{kg}^2$.

Now let the radius of the star be R, the mass enclosed inside a sphere of radius r be M_r, and the mass enclosed in a spherical shell of thickness dr and volume $dV = 4r^2\pi dr$ be dm. The density of the sphere can be obtained as

$$M = \varrho \cdot \frac{4}{3}R^3\pi \Rightarrow \varrho = \frac{3M}{4R^3\pi}, \tag{10}$$

where ϱ is the density of the sphere and M is its total mass. Hence,

$$M_r = \varrho\frac{4}{3}r^3\pi = M\frac{r^3}{R^3},$$
$$dm = \varrho 4r^2\pi dr = \frac{3Mr^2dr}{R^3}.$$

From this,

$$dE_g = -G\frac{M_r dm}{r}. \tag{11}$$

The total contribution can be obtained by integrating over the entire sphere:

$$E_g = -G\int_0^R M\frac{r^3}{R^3}\frac{3Mr^2}{R^3}\frac{1}{r}\,dr = -\frac{3GM^2}{R^6}\int_0^R r^4\,dr = -\frac{3}{5}\frac{GM^2}{R}. \tag{12}$$

An important mathematical observation is that in this case we formulated the integral directly in terms of the radial coordinate, without performing a coordinate transformation; therefore, no Jacobian determinant has to be taken into account.

Using $M = Am_n$, where m_n is the neutron mass, together with $R = r_0A^{1/3}$, the energy originating from the gravitational field becomes

$$E_g = -\frac{3}{5}\frac{Gm_n^2}{r_0}\frac{A^2}{A^{1/3}} = -e_g A^{5/3}, \tag{13}$$

where we used the fact that all prefactors in front of the expression containing the mass number are constants. A further simplification is that at such scales the surface correction term can be neglected compared to the volume term. Thus, for the binding energy:

$$E(A) = -e_V A + e_a A - e_g A^{5/3} < 0, \tag{14}$$

where we used that the total energy is negative for a bound state. Rearranging yields

$$\left(\frac{e_a - e_V}{e_g}\right) < A. \tag{15}$$

It can be calculated that $e_g \approx 9.3 \cdot 10^{-50}$ J. From this one obtains $A > 5.24 \cdot 10^{55}$. The minimum mass of such an object can then be determined using $m_n = 1.67 \cdot 10^{-27}$ kg:

$$M = m_n A = 1.67 \cdot 10^{-27}\,\text{kg} \cdot 5.24 \cdot 10^{55} = 8.76 \cdot 10^{28}\,\text{kg}. \tag{16}$$

This is approximately one twentieth of the mass of our Sun.

3. Problem 7

***On the Cassini spacecraft traveling toward Saturn, the heat generated by the α-decay of the isotope ^{238}Pu is used to produce electrical energy. The spacecraft requires 530 W of electrical power to operate its equipment in orbit around Saturn. What is the minimum mass of ^{238}Pu used in constructing the power source?**

Data:

- **The half-life of ^{238}Pu is 88 years.**
- **This isotope undergoes pure α-decay, releasing 5.6 MeV of energy per decay, 90% of which is converted into heat!**
- **The heat can be converted into electrical energy with an efficiency of 7%.**

Solution: In the Pu-238 decay chain there happens 8 alpha decay. The half-life of the parent nucleus (Pu-238, T=88 years) is many orders of magnitude shorter than that of the daughter nuclei. Let us suppose that the power is constantly needed for at most 5 years. During that amount of time the decays of the daughter nuclei are not significant, their activity can be neglected. Now let us calculate how the activity A of Pu-238 changes over 5 years time:

$$A = A_0 \times e^{-\lambda t} = A_0 e^{-5/88} \approx 0.96 A_0. \tag{17}$$

Hence, it can be concluded that the activity is approximately constant.

What we need is the efficient energy of the decay over 1 second to be 530 J.

What is the energy released in 1 second? It is the alpha decay energy (E = 5.6 MeV) of a single nucleus multiplied by the number of decays occurring in 1 second. The number of decays in 1 second is given by the activity of Pu-238 ($A = \lambda N$). Hence, we have the following equation:

$$530 J/s = E \cdot 0.9 \cdot 0.07 \cdot (\ln 2/T) \cdot (m/M) \cdot 6 \cdot 10^{23}. \tag{18}$$

Knowing the half-life and molar mass (238 g / mol), the mass m needed for the electric equipment to operate is approximately $m = 14.8$ kg.

4. Problem 9

***The activity concentration of tritium ([c]=Bq/l) in the water of the Danube River is on average 6 Bq per liter. Calculate what fraction of the hydrogen isotopes present in the water is tritium.**

Solution: Let us denote the activity per unit volume as the *activity concentration*, $C = A/V = 6\,\mathrm{Bq/l}$. The half-life of tritium is $T_{1/2} = 12.32\,\mathrm{y}$, meaning its decay constant is $\lambda = 1.784 \cdot 10^{-9}\,\mathrm{s}^{-1}$. From this, the number of tritium atoms per liter is:

$$\tilde{N}_\mathrm{T} = \frac{C}{\lambda} = 3.36 \cdot 10^9\,\mathrm{atoms/l}. \tag{19}$$

Given that the density of water is approximately $\varrho = 1000\,\mathrm{g/l}$, and the molar mass of (light) water is $M = 18\,\mathrm{g/mol}$, we can calculate the molar concentration of water as:

$$\tilde{n} = \frac{\varrho}{M} = 55.56\,\mathrm{mol/l}.$$

From this it follows that in 1 liter of water, the number of hydrogen atoms is:

$$\tilde{N}_\mathrm{H} = 2 \cdot \tilde{n} N_A = 6.67 \cdot 10^{27}\,\mathrm{atoms/l},$$

where N_A is Avogadro's number, and the factor of 2 arises from the fact that each water molecule contains two hydrogen atoms.

From this, the relative abundance of tritium is:

$$\kappa_\mathrm{T} = \frac{\tilde{N}_\mathrm{T}}{\tilde{N}_\mathrm{H}} = \frac{3.36 \cdot 10^9}{6.67 \cdot 10^{27}} = 5.038 \cdot 10^{-17}. \tag{20}$$

5. Problem 10

***The decay chain of ^{232}Th (Z = 90, half-life = 14.05 billion years) ends with the stable isotope ^{208}Pb (Z = 82). How many alpha decays occur in the full decay chain?**

Solution:

The decay chain includes alpha and beta decays (we ignore gamma decays in the practice sessions). Among the different types of radioactive decay—excluding spontaneous fission—only alpha decay alters the mass number. When a nucleus in the decay chain undergoes alpha decay, its mass number (A) decreases by 4, and its atomic number (Z) decreases by 2.

Let us examine the mass number and atomic number of both the parent and daughter nuclei. The change in the mass number is: 232-208=24. Since only alpha decay changes this number and it is reduced by 4 each time, we conclude that the number of alpha decays is $n_\alpha = 232 - 208/4 = 6$.

In the case of beta decay, the mass number remains unchanged (an isobaric transition); however, the atomic number either decreases by 1 or increases by 1 depending on the type of beta decay. Alpha decay decreases Z by 2 each time, resulting in: $Z' = 90 - (2 \cdot n_\alpha) = 78$, however we know that the atomic number (Z) of the final element in the decay series is Z=82, which indicates that an additional $n_\beta = 82 - 78 = 4$ negative beta decay must occure. As a result: $n = n_\alpha + n_\beta = 10$. *The total activity of the decay series arises from all 10 decays.*

6. Problem 13

***The natural uranium found on Earth typically contains 0.71% of the isotope ^{235}U. However, in samples from a uranium mine located in Oklo, a region in the Central African country**

of Gabon, researchers were surprised to find that the ^{235}U concentration was only 0.44%. Additionally, fission products of uranium (such as ^{143}Nd) were detected in the surrounding area of the site. Based on these findings, it was hypothesized that a „natural nuclear reactor" might have operated there in the distant past, with groundwater acting as a moderator by slowing down the neutrons. However, in the case of moderation with natural (light) water, a chain reaction can only be sustained if the ^{235}U concentration is at least 3%.

Question: How long ago might the „reactor phase" have ended?

Data: The half-life of ^{235}U is 703.8 million years, and the half-life of ^{238}U is 4.47 billion years.

Solution: One of the major limiting factors of nuclear power generation (and weapons development) is that the uranium isotope ^{238}U is irrelevant in terms of fission, yet it constitutes the vast majority of natural uranium (which is why fuel is typically enriched). Furthermore, over time, the fraction of ^{235}U continuously decreases because it decays faster (see half-lives). In this „natural reactor", the lower current proportion of ^{235}U is due to the fact that it was consumed more quickly during the reactor phase, reaching the 3% limit earlier than uranium found elsewhere on Earth (where such a natural reactor did not operate because there was nothing to moderate the neutrons).

Let us define time zero as the moment when the reactor shut down, i.e., when the proportion of the ^{235}U isotope was 3%, meaning

$$0.03 = \frac{N_0^{235}}{N_0^{235} + N_0^{238}} \rightarrow N_0^{238} = 32.33 N_0^{235}. \tag{21}$$

We also know that the current relative ratio of the two isotopes is 0.44%, thus:

$$0.0044 = \frac{N^{235}(t)}{N^{235}(t) + N^{238}(t)} = \frac{N_0^{235} e^{-\lambda_{235} t}}{N_0^{235} e^{-\lambda_{235} t} + N_0^{238} e^{-\lambda_{238} t}}. \tag{22}$$

Substituting (21) into this expression gives:

$$0.0044 = \frac{1}{1 + 32.33 \exp\left(\ln(2)\left(\frac{1}{T_{1/2}^{235}} - \frac{1}{T_{1/2}^{238}}\right) t\right)}. \tag{23}$$

Rearranging this expression, we get:

$$t = \frac{\ln(7)}{\ln(2)} \cdot \frac{1}{\frac{1}{T_{1/2}^{235}} - \frac{1}{T_{1/2}^{238}}} = 2.345\,\text{by}, \tag{24}$$

which means this reactor shut down approximately two and a half billion years ago.

7. Problem 14

Among the radioactive nuclides released during the Chernobyl accident were ^{131}I ($\mathbf{T_{1/2} = 8}$ days) and ^{137}Cs ($\mathbf{T_{1/2} = 30.23}$ years). In the operating reactor, approximately five times more ^{137}Cs was produced than ^{131}I at the time of the accident.

a) Assuming the entire inventory was released into the environment, after how much time did the activity of ^{137}Cs become five times greater than that of ^{131}I?

b) During this time period (measured from the time of release), by what factor did the total number of decayed ^{131}I atoms exceed that of ^{137}Cs?

Solution: We know that $N_0^{\text{Cs}} = 5 \cdot N_0^{\text{I}}$. Then, the ratio of activities is:

$$\frac{A_0^{\text{I}}}{A_0^{\text{Cs}}} = \frac{\lambda_{\text{I}} N_0^{\text{I}}}{\lambda_{\text{Cs}} N_0^{\text{Cs}}} = \frac{1}{5} \frac{T_{1/2}^{\text{Cs}}}{T_{1/2}^{\text{I}}} = 275.5, \tag{25}$$

which means that even though the amount of cesium was larger, the main radiation exposure came from iodine.

We are looking for the point in time when the activity of cesium became five times greater than the activity of iodine, i.e.,

$$\frac{1}{5} = \frac{A_{\mathrm{I}}(t)}{A_{\mathrm{Cs}}(t)} = \frac{A_0^{\mathrm{I}} e^{-\lambda_{\mathrm{I}} t}}{A_0^{\mathrm{Cs}} e^{-\lambda_{\mathrm{Cs}} t}} = 275.5 \cdot e^{-(\lambda_{\mathrm{I}} - \lambda_{\mathrm{Cs}})t}, \tag{26}$$

thus,

$$t = \frac{1}{\frac{1}{T_{1/2}^{\mathrm{I}}} - \frac{1}{T_{1/2}^{\mathrm{Cs}}}} \cdot \frac{\ln(5 \cdot 275.5)}{\ln(2)} = 83.48\,\mathrm{d}. \tag{27}$$

Considering that the accident occurred on April 26, 1986, at 1:24 AM, in our model the 5x cesium-to-iodine activity ratio was reached in the early afternoon of July 18, 1986.

The number of decayed nuclei during this time is:

$$\Delta N_{\mathrm{I}}(t) = N_0^{\mathrm{I}} - N_{\mathrm{I}}(t) = N_0^{\mathrm{I}} \left(1 - e^{-\lambda_{\mathrm{I}} t}\right), \tag{28}$$

$$\Delta N_{\mathrm{Cs}}(t) = N_0^{\mathrm{Cs}} - N_{\mathrm{Cs}}(t) = N_0^{\mathrm{Cs}} \left(1 - e^{-\lambda_{\mathrm{Cs}} t}\right), \tag{29}$$

so their ratio is:

$$\frac{\Delta N_{\mathrm{I}}(t)}{\Delta N_{\mathrm{Cs}}(t)} = \frac{N_0^{\mathrm{I}}}{N_0^{\mathrm{Cs}}} \cdot \frac{1 - e^{-\lambda_{\mathrm{I}} t}}{1 - e^{-\lambda_{\mathrm{Cs}} t}} = \frac{1}{5} \cdot \frac{1 - 2^{-t/T_{1/2}^{\mathrm{I}}}}{1 - 2^{-t/T_{1/2}^{\mathrm{Cs}}}} = 38.16, \tag{30}$$

which means that nearly 40 times as many iodine nuclei decayed as cesium nuclei.

We may also note that the number of decayed nuclei over an interval is given by:

$$\Delta N(t, t_0) = N(t_0) - N(t) = \int_{t_0}^{t} \left(-\frac{dN(t)}{dt}\right) dt = \int_{t_0}^{t} A(t)\, dt = A(t_0) \int_{t_0}^{t} e^{-\lambda t}\, dt = \frac{A(t_0) e^{-\lambda t_0}}{\lambda} \left(1 - e^{-\lambda (t - t_0)}\right) = N_0 \left(1 - e^{-\lambda \Delta t}\right). \tag{31}$$

8. Problem 17

Prove that the fusion reaction

$$^2\mathbf{H} + {}^2\mathbf{H} \to {}^4\mathbf{He}$$

cannot occur!

Data:

$$M(^2\mathrm{H}) = 2.014101\,\mathrm{u}, \quad M(^4\mathrm{He}) = 4.002603\,\mathrm{u}$$

Note: Instead, the reaction

$$^2\mathrm{H} + {}^3\mathrm{H} \to {}^4\mathrm{He} + n$$

is used for energy generation through fusion.

Solution: During the process, energy is released:

$$Q = \left(2M_{^2_1\mathrm{H}} - M_{^4_2\mathrm{He}}\right) c^2 = 0.025599\,c^2 > 0. \tag{32}$$

This energy should be carried away by the resulting particle, the $^4_2$He, in the form of kinetic energy, since it is known that $^4_2$He has no accessible excited states. Let us examine the process in the center-of-mass frame! In this frame, the total momentum is zero in the initial state (the two deuterons move toward each other with momenta of equal magnitude but opposite direction), so in the final state the produced $^4_2$He must also have zero momentum.

However, this is impossible, since a $^4_2$He nucleus with zero momentum cannot have kinetic energy greater than zero!